

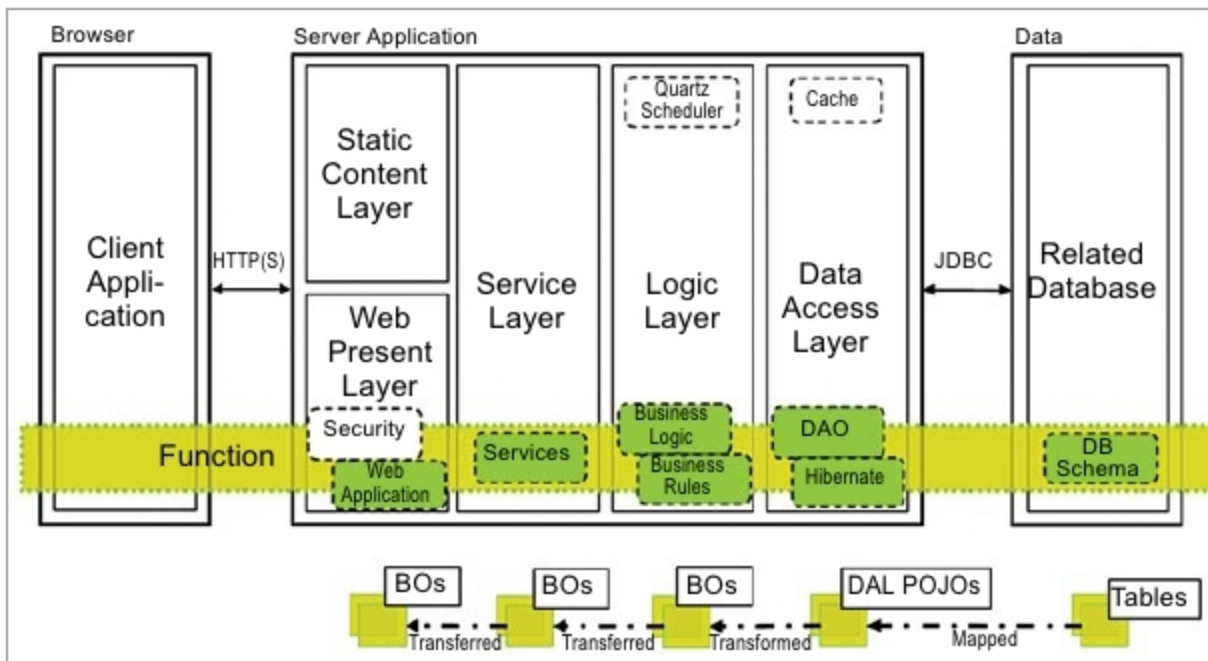


Fig. 1: Software-centric service architecture (Credit: <https://applicationarchitecture.wordpress.com>)

including managing transactions, tracking leads, scheduling events, auditing and so on.

According to Sameer Redij, chief revenue officer and managing partner, EvolutionCo Digital & Interactive Consultancy, “Digital technologies are disrupting the way traditional business models operate. Customers are demanding transformational experiences that will redefine their organisation and its relationship with its ecosystem (clients, partners, employees, etc). Whether it is improving employee productivity or building customer experiences, more and more organisations are relying on the Web and a mobile-based ecosystem. Software solutions are now redefining businesses and processes. There are ample business cases like online marketplaces that have been innovative and transformational.”

Web-based solutions for businesses

SaaS applications for fundamental business technologies are email management, sales management, customer relationship management (CRM), financial management, human resource management, billing and collaboration, among others. Leading SaaS providers include Oracle, Salesforce, SAP, Microsoft, Intuit and many others.

SaaS offers pay-as-you-go services where one needs to pay the amount only for the time of use of an app or by subscription chosen. All underlying infrastructure, app software, app data

and middleware are located on the service provider’s data centre. The service provider manages the hardware and software.

An appropriate service agreement ensures availability and security of the app and data. One can access emails and stored messages from a Web browser on any computer or Internet-connected device.

Examples of software-centric business models

Uber is a classic example of a software-centric business model. There are similar examples of cloud-based service providers like Amazon and Zomato, which are breaking traditional boundaries and creating new business avenues. The emphasis of these businesses is on finding opportunities and problems, and providing digital solutions that address these in an experiential manner.

EvolutionCo is a digital transformation and consulting-approach-based organisation. It provides custom software solutions (business applications) to clients across the business spectrum, right from human resource, finance, sales and marketing to operations, CRMs, project management software, and a suite of different portal and mobile-based solutions.

Advantages of SaaS

There is no hardware cost involved, as processing power is supplied by the cloud provider. Applications are ready to use once the user subscribes, with no initial setup costs. There is

no need to install new software and hardware to access additional storage or on-demand services.

Updates are often deployed automatically by the software provider. Applications can be altered to suit individual needs. Users can run most SaaS apps directly from their Web browser. SaaS makes even sophisticated enterprise applications, such as ERP and CRM, affordable for organisations that lack the resources like infrastructure and software.

SaaS makes it easy to mobilise the workforce because users can access SaaS apps and data from any Internet-connected computer or mobile device. No data is lost if a user’s computer or device fails, as app data is stored in the cloud. Security of data is ensured by the service provider regardless of the type of device consuming it.

SaaS offers high scalability. It gives access of services to customers on demand. The SaaS provider automatically performs updates and patch management for all software applications.

Security in SaaS model

When someone on the outside maintains an organisation’s critical data, security is crucial. Data security is independent of the location of the server or the user. To ensure security, it is important for a data centre to undergo stringent security procedures of SAS70 type II audits.

SLAs set system reliability standards and parameters for issues such as data ownership, security requirements and maintenance. Users should ensure the clause in SLAs regarding the state of ownership of data.

Most SaaS contracts provide access to data and back it up locally as needed. However, contracts should also include type of format and availability for data, in addition to its migration for an appropriate fee.

Software-based solution for devices

Javier Barabas, IBM cloud and cognitive technical leader, says, “As far as software-centric technologies for devices are concerned, all IoT technologies will fit in this space.”

Redij adds, “Innovative software